PRACTICAL 9

Aim : Study of firewall in providing network security.

Input :

Introduction :

A firewall is a network security system that monitors and controls incoming and outgoing network traffic based on predefined security rules. It acts as a barrier between a trusted internal network and an untrusted network, such as the Internet.

Objectives of a Firewall :

- To prevent unauthorized access to a network.
- To protect computers and network resources from malicious traffic.
- To monitor and control network communication.
- To block suspicious or unwanted connections.
- To enforce an organization's security policies.

How a Firewall Provides Network Security :

A firewall examines network packets and decides whether to allow, reject, or block them according to configured rules. For example, it can allow web traffic through HTTP/HTTPS while blocking unauthorized ports or connections.

The basic process is:

Internet → Firewall → Internal Network

The firewall checks traffic before it reaches protected systems, thereby reducing the risk of unauthorized access and attacks.

Types of Firewalls :

1. **Packet-Filtering Firewall** – Examines packet information such as source/destination IP addresses, ports, and protocols.
2. **Stateful Firewall** – Tracks active connections and makes decisions based on the state of network sessions.
3. **Proxy Firewall** – Acts as an intermediary between users and external networks.
4. **Next-Generation Firewall (NGFW)** – Provides advanced inspection, application control, intrusion prevention, and other security capabilities.
5. **Host-Based Firewall** – Runs directly on an individual computer or server.

Advantages :

- Protects against unauthorized network access.
- Controls network traffic.
- Helps prevent certain types of malware and attacks.
- Provides logging and monitoring of network activity.
- Supports enforcement of security policies.
- Can help isolate sensitive parts of a network.

Limitations :

- A firewall cannot protect against every type of attack.
- It cannot fully prevent attacks caused by compromised or careless users.
- Incorrect firewall configuration can create security vulnerabilities.
- Encrypted traffic may require additional security mechanisms for effective inspection.
- Firewalls should be used together with antivirus, intrusion detection/prevention, authentication, and other security measures.

Conclusion :

A firewall is an important component of network security because it creates a controlled boundary between trusted and untrusted networks. By filtering and monitoring traffic according to security rules, it helps protect systems and data from unauthorized access. However, a firewall is not a complete security solution and should be combined with other security controls for effective protection.